

Conservative Road-Scene Risk Assessment via Fail-Aware Multi-Model Fusion

Pham Hoang Bach · Vu Gia Tien · Trinh Quang Tien · Cao Van Mai

Faculty of Artificial Intelligence, FPT University, Hanoi, Vietnam

ICoCSETI 2026 · Jakarta, Indonesia

OPENING

One image, three kinds of uncertainty

Detector misses

A missed vehicle or pedestrian silently turns into an empty road, especially at night and in rain.

Depth has no scale

Monocular depth orders the scene near to far, but gives no meters and becomes unstable on dark, blurred or occluded objects.

Path is ambiguous

A box does not say whether the object is in the ego lane, at its edge, or parked beside the road.

Goal: weak or conflicting evidence may add caution, but must never produce a confident CLEAR or an unsupported STOP.

OPENING

What this work contributes

Interpretable fusion

YOLO11 objects related to a HybridNets ego corridor and Depth Anything V2 relative depth; weak evidence can add caution but cannot justify a severe advisory.

Multi-region depth reliability

Each object is labeled STABLE, CHECK or UNCERTAIN from the agreement of three mask regions, and validated against annotations.

Severity gates and fallback

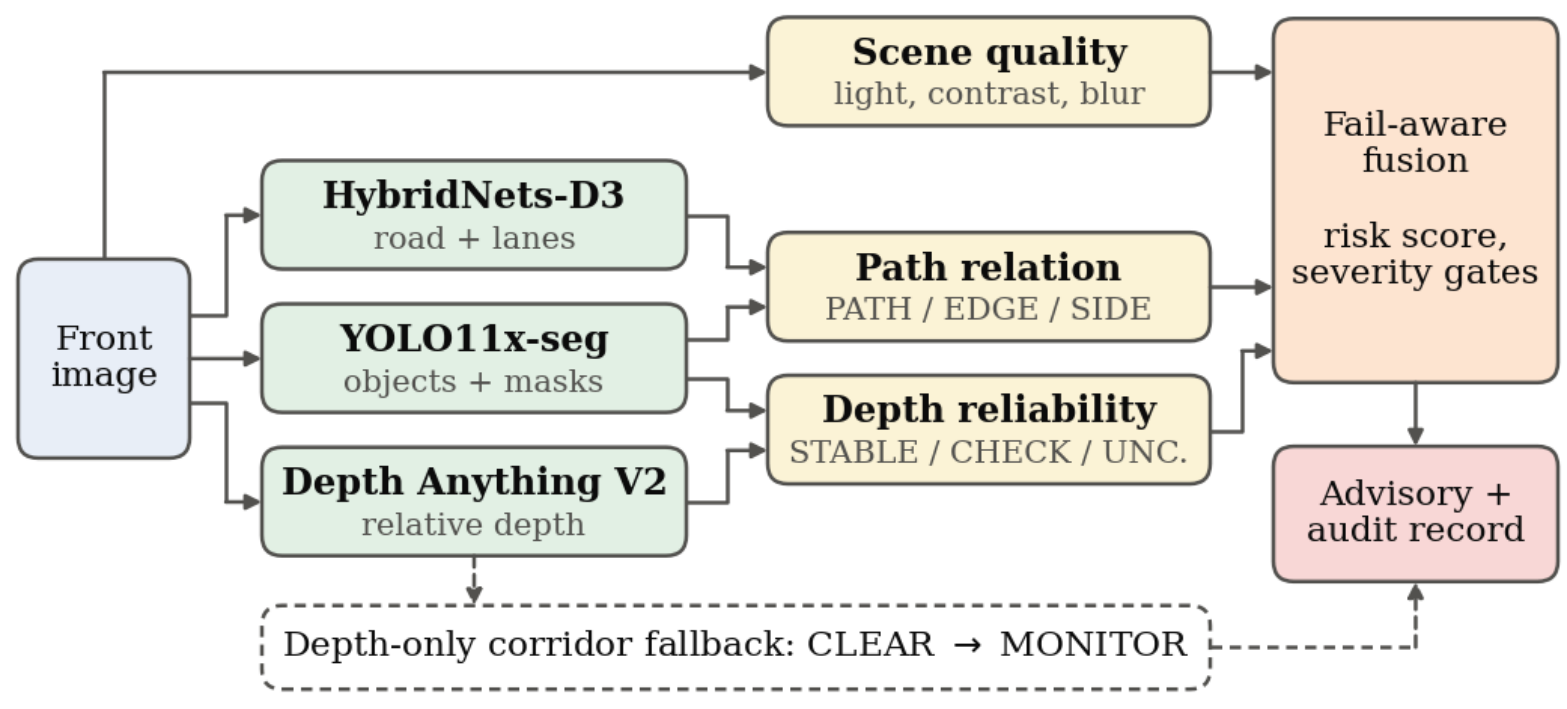
Caps on unsupported BRAKE/STOP and a depth-only fallback for detector-empty images, measured with ground-truth lane occupancy.

Reproducible evaluation

9,500 BDD100K validation images, released image IDs, cached-replay ablations, confidence intervals and exact McNemar tests.

METHOD

Three frozen models, one fail-aware fusion layer



No training

All three networks are used as released; only the fusion layer is ours.

Relative depth only

Depth orders objects within one image; it is never read as meters.

Audit record

Every object keeps its confidences, depth state, path relation and advisory.

YOLO11x-seg 960 px, 4-pass TTA · HybridNets-D3 640 px · Depth Anything V2 ViT-L 518 px

METHOD

Scene quality and a relative depth index

Scene quality q_s (from the gray image)

q_s = 0.40 q_illum + 0.30 q_contrast + 0.30 q_sharp

median brightness m, 10–90 percentile range R, Laplacian variance v

Flags: low light m < 45, low contrast R < 55, blur v < 60, glare

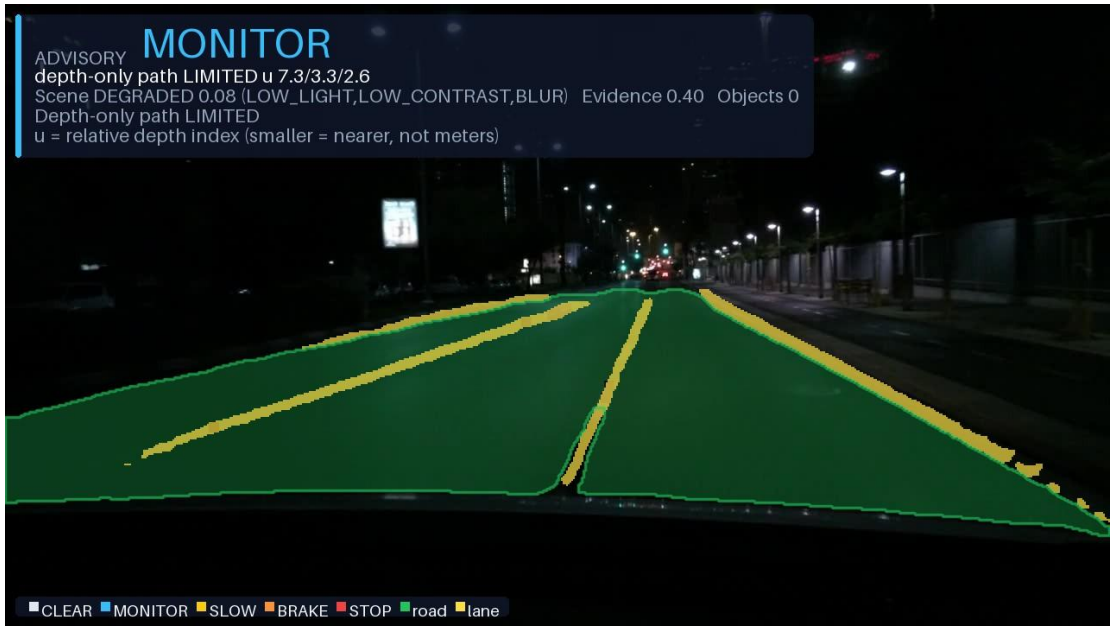
DEGRADED if q_s < 0.35 or dark and blurred; LIMITED if q_s < 0.58 or any flag

Relative depth index u (from Depth Anything V2)

u = clip(1000 / d, 0.1, 999)

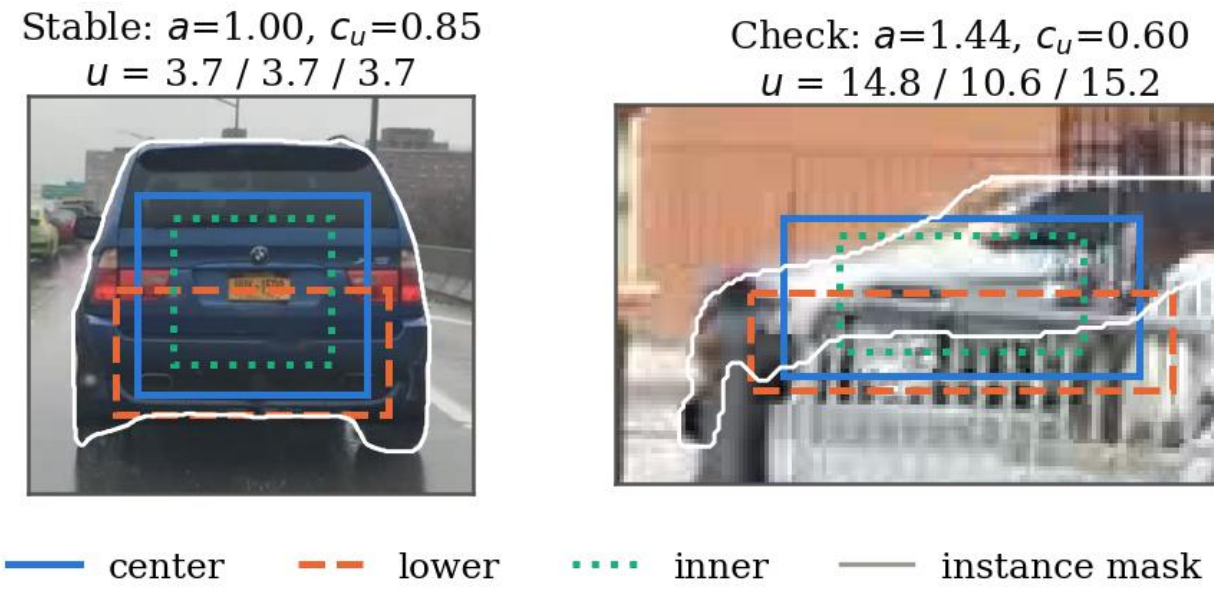
smaller u = nearer surface; the network output d has unknown scale and shift

used only to order objects within one image, never as meters



DEGRADED night scene, no object detected. Scene quality 0.08 (low light, low contrast, blur); the depth-only corridor check is LIMITED, so CLEAR is raised to MONITOR.

Do the parts of an object agree on its depth?



Left: the three regions agree (a = 1.00) → STABLE. Right: the mask leaks into the background (a = 1.44) → CHECK.

How the state is set

- Relative depth u is sampled in the center, lower-body and inner-core regions of the mask
- Disagreement ratio a = max u_r / min u_r
- Reliability c_u combines a, the spread of the samples and the support

STABLE	c_u ≥ 0.68 and a ≤ 1.35
CHECK	c_u ≥ 0.35 and a ≤ 1.80
UNCERTAIN	otherwise

METHOD

From evidence to a conservative advisory

CLEAR	MONITOR	SLOW	BRAKE	STOP
risk < 24	risk ≥ 24	risk ≥ 40	risk ≥ 60	risk ≥ 78



Path relation

PATH adds +32 and EDGE +15 to the risk score. A SIDE object can raise at most MONITOR (SLOW for a close vulnerable road user).



Severity gates

STOP needs a close, STABLE, in-path object with detector ≥ 0.55 and evidence ≥ 0.48. Weak evidence caps BRAKE and SLOW.



Depth-only fallback

No object detected? If the depth along the corridor is not consistent, CLEAR is raised to MONITOR. It never creates an object.

EVALUATION

Evaluated against BDD100K ground truth

Split	Images	Use
Test	**9,500**	every headline number
Dev	500	priors, sensitivity, operating curve only
Hard set	100	dark, low contrast, blur, glare, complex
1k subset	1,000	perturbation stability
val_rest	8,400	held-out operating-point check
Train	69,863	supplementary (HybridNets saw it)

Image IDs and SHA-256 hashes of every split are released.

Definitions, fixed before any result

Occupied image
a relevant road user (≥ 5% of image height) stands in the annotated ego lane

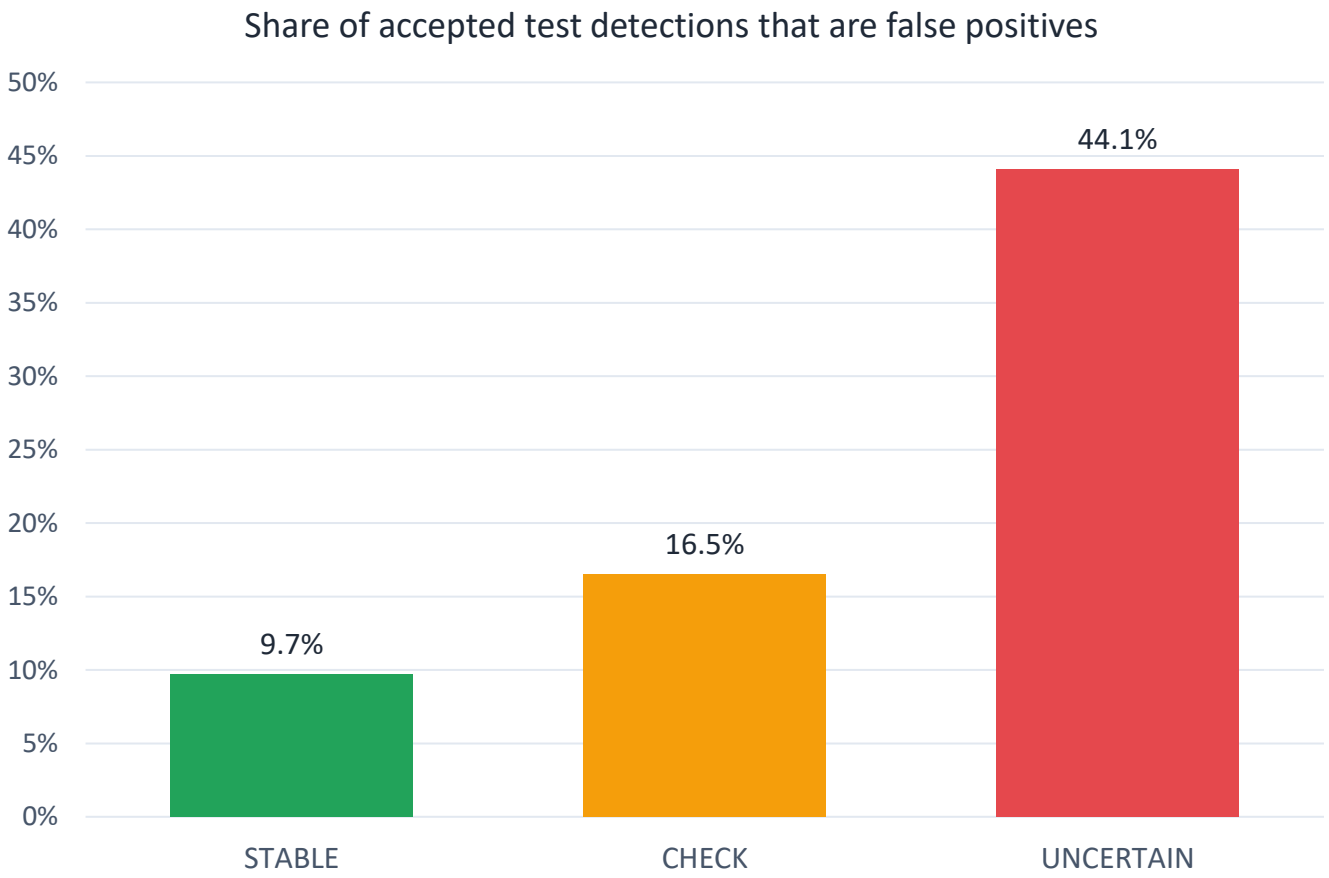
Unsafe CLEAR
CLEAR advisory on an occupied image

Over-caution
SLOW or higher while no road user is in the ego lane

Statistics
Wilson intervals, image-cluster bootstrap (10,000), exact McNemar

RESULTS

Reliability states carry real signal



84.9%

of the 75,819 accepted test detections are STABLE

1.8×

non-STABLE estimates are false positives 1.8 times as often (17.4% vs 9.7%)

69%

of CHECK objects are truncated by the image border, against 2.5% of STABLE ones

RESULTS

All headline numbers at a glance

Measure	Hard set (100)	Test split (9,500)
Accepted detections	505	**75,819**
STABLE / CHECK / UNCERTAIN	437 / 67 / 1	64,334 / 11,104 / 354
Stable ratio [95% CI]	86.5 [83.3, 89.2]	**84.9 [84.6, 85.1]**
Detection precision (GT)	86.3	89.1
Recall of relevant road users	72.2	78.9
PATH precision / recall	15.8 / 56.8	9.2 / 53.8
Corridor IoU with GT ego lane	0.242	0.240
Unsafe CLEAR (fallback on)	0 / 30	**11 / 3,250 (0.3%)**
Unsafe CLEAR (fallback off)	1 / 30	15 / 3,250 (0.5%)
Over-caution (free ego lane)	34 / 67 (50.7%)	**4,799 / 5,844 (82.1%)**
Unsupported BRAKE / STOP	31 / 47	4,179 / 6,188
Detector-empty / fallback fired	15 / 15	295 / 236

How to read it

Percentages unless counts are shown; brackets are 95% confidence intervals.

The hard set is a stress test of 100 dark, low-contrast, blurred, glare and complex images.

Over-caution is an upper bound: an adjacent-lane hazard can justify some of these advisories.

RESULTS

Safe, at a measured cost

Safety

11 / 3,250

occupied test images received an unsafe CLEAR (0.34%, upper bound 0.6%); 0 of 30 on the hard set

6 of the 11: object detected but placed beside the corridor. 5: missed by the detector (4 at night).

The price

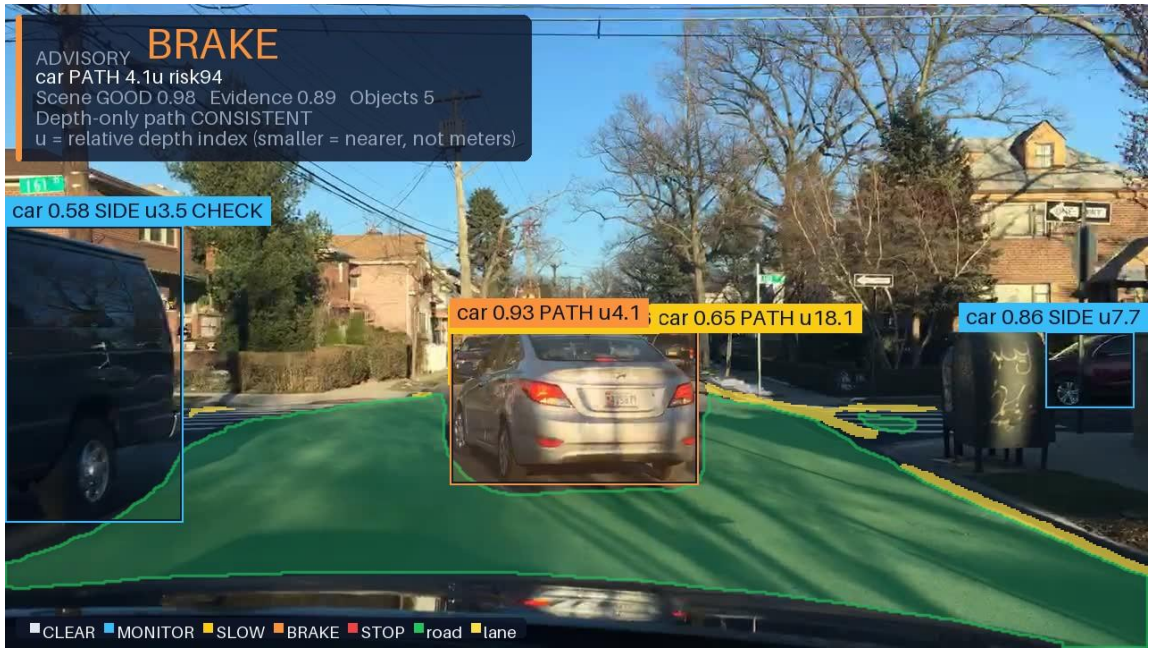
82.1%
over-caution on free-lane images

2.1%
of test images receive CLEAR

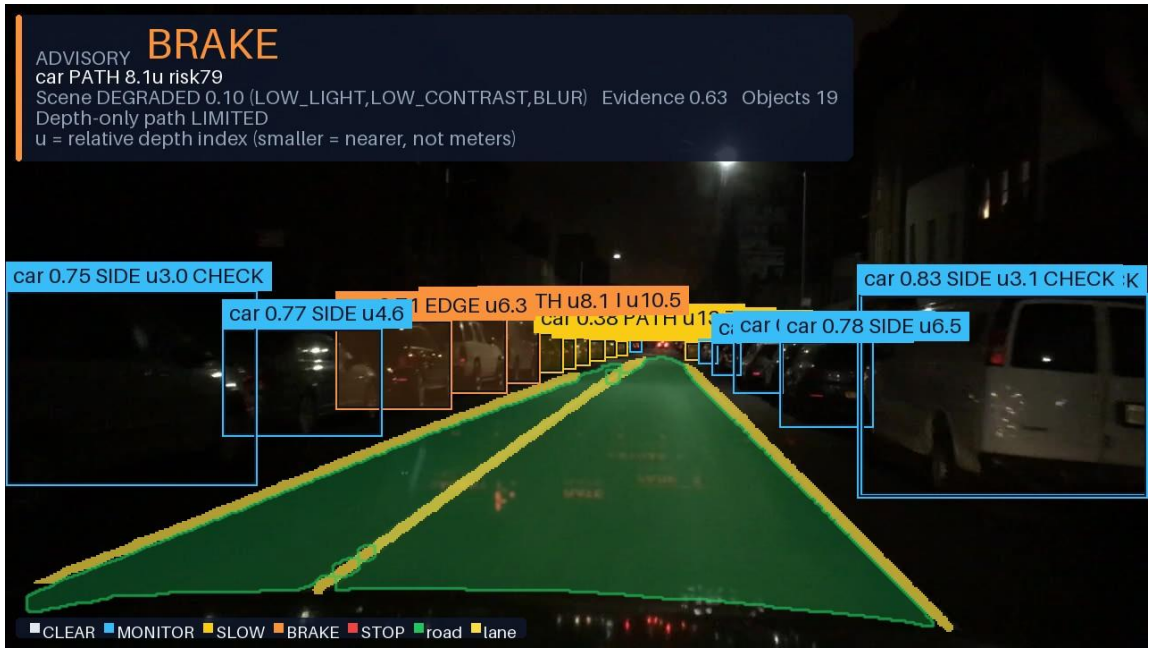
9.2%
PATH precision against the ego lane

Cause: the ego corridor. The fixed perspective band overlaps the annotated ego lane with IoU 0.24, while HybridNets' own road mask reaches 0.83. Roadside and distant objects are labeled PATH.

What the system shows



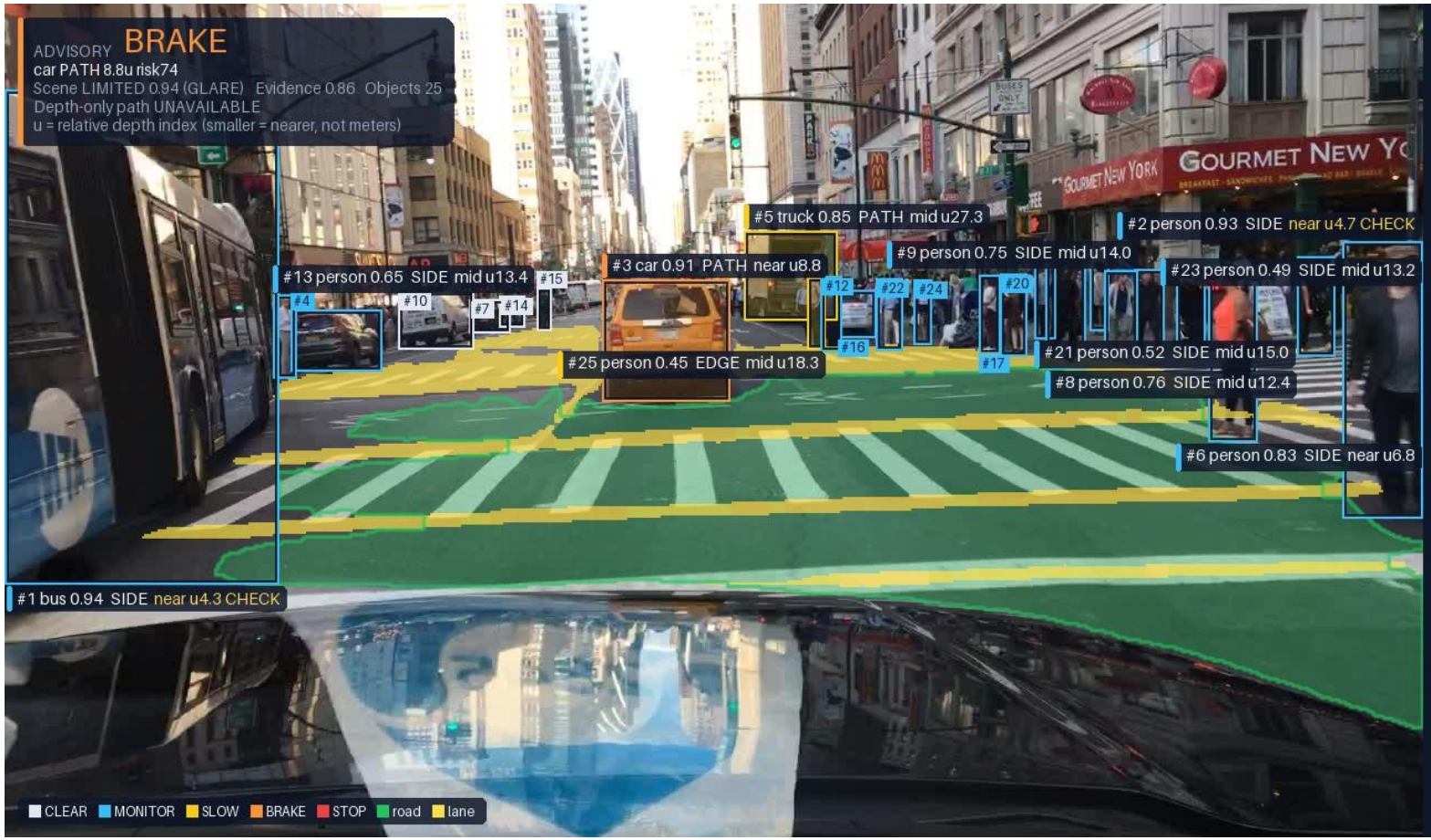
Daytime. The car ahead is PATH and raises BRAKE; parked cars are SIDE and stay at MONITOR.



Night, degraded scene. The gates keep the advisory at BRAKE; depth that is not STABLE is labeled CHECK.

Every box: class · confidence · PATH/EDGE/SIDE · u (relative depth, smaller = nearer, not meters) · CHECK if depth not STABLE

Every detection, with its numbers



All 25 detected objects

#	class	conf	path	range	u	depth	advisory
1	bus	0.94	SIDE	near	4.3	Check	MONITOR
2	person	0.93	SIDE	near	4.7	Check	MONITOR
3	car	0.91	PATH	near	8.8	Stable	BRAKE
4	car	0.87	SIDE	mid	14.8	Stable	MONITOR
5	truck	0.85	PATH	mid	27.3	Stable	SLOW
6	person	0.83	SIDE	near	6.8	Stable	MONITOR
7	car	0.78	SIDE	far	30.6	Stable	CLEAR
8	person	0.76	SIDE	mid	12.4	Stable	MONITOR
9	person	0.75	SIDE	mid	14.0	Stable	MONITOR
10	truck	0.68	SIDE	mid	22.8	Stable	CLEAR
11	person	0.66	SIDE	mid	13.4	Stable	MONITOR
12	person	0.66	SIDE	mid	16.5	Stable	MONITOR
13	person	0.65	SIDE	mid	13.4	Stable	MONITOR
14	car	0.63	SIDE	far	37.1	Stable	CLEAR
15	person	0.63	SIDE	far	32.3	Stable	CLEAR
16	car	0.62	SIDE	mid	16.7	Stable	MONITOR
17	person	0.60	SIDE	mid	14.9	Stable	MONITOR
18	person	0.60	SIDE	mid	16.1	Stable	MONITOR
19	person	0.59	SIDE	mid	12.4	Stable	MONITOR
20	person	0.54	SIDE	mid	14.3	Stable	MONITOR
21	person	0.52	SIDE	mid	15.0	Stable	MONITOR
22	person	0.51	SIDE	mid	16.2	Stable	MONITOR
23	person	0.49	SIDE	mid	13.2	Stable	MONITOR
24	person	0.46	SIDE	mid	17.5	Stable	MONITOR
25	person	0.45	EDGE	mid	18.3	Stable	SLOW

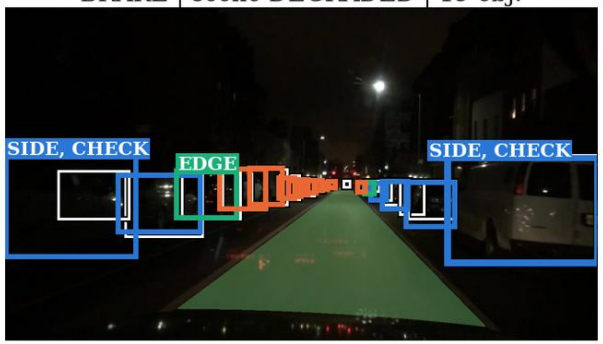
Each box carries its number; the table lists all 25 objects. Detected here: 21 of the 34 relevant road users in the annotations. The others have no box because the detector missed them (recall 78.9% on the test split), not because the display hides them.

Six hard-set cases, compared with the annotations

(a) Dense scene: correct PATH objects
BRAKE | scene LIMITED | 25 obj.



(b) Degraded night: gates cap STOP
BRAKE | scene DEGRADED | 19 obj.



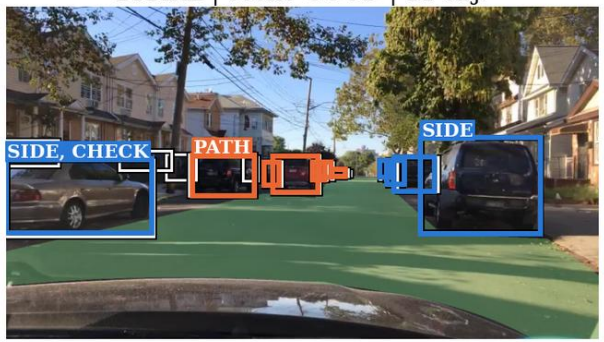
(c) Detector-empty: fallback MONITOR
MONITOR | scene DEGRADED | 0 obj.



(d) Failure: distant parked car as PATH
BRAKE | scene GOOD | 13 obj.



(e) Disagreement: annotated lane spans the street
BRAKE | scene GOOD | 11 obj.



(f) Truncated object kept at CHECK
BRAKE | scene GOOD | 14 obj.



PATH EDGE SIDE BDD100K box (white/black) GT ego lane (direct)

Colored boxes: system output by path relation (PATH, EDGE, SIDE; CHECK = depth not STABLE) · white boxes: BDD100K annotations · green: annotated ego lane

(a–c) intended behaviour
correct PATH objects; gates cap STOP at night; fallback raises MONITOR

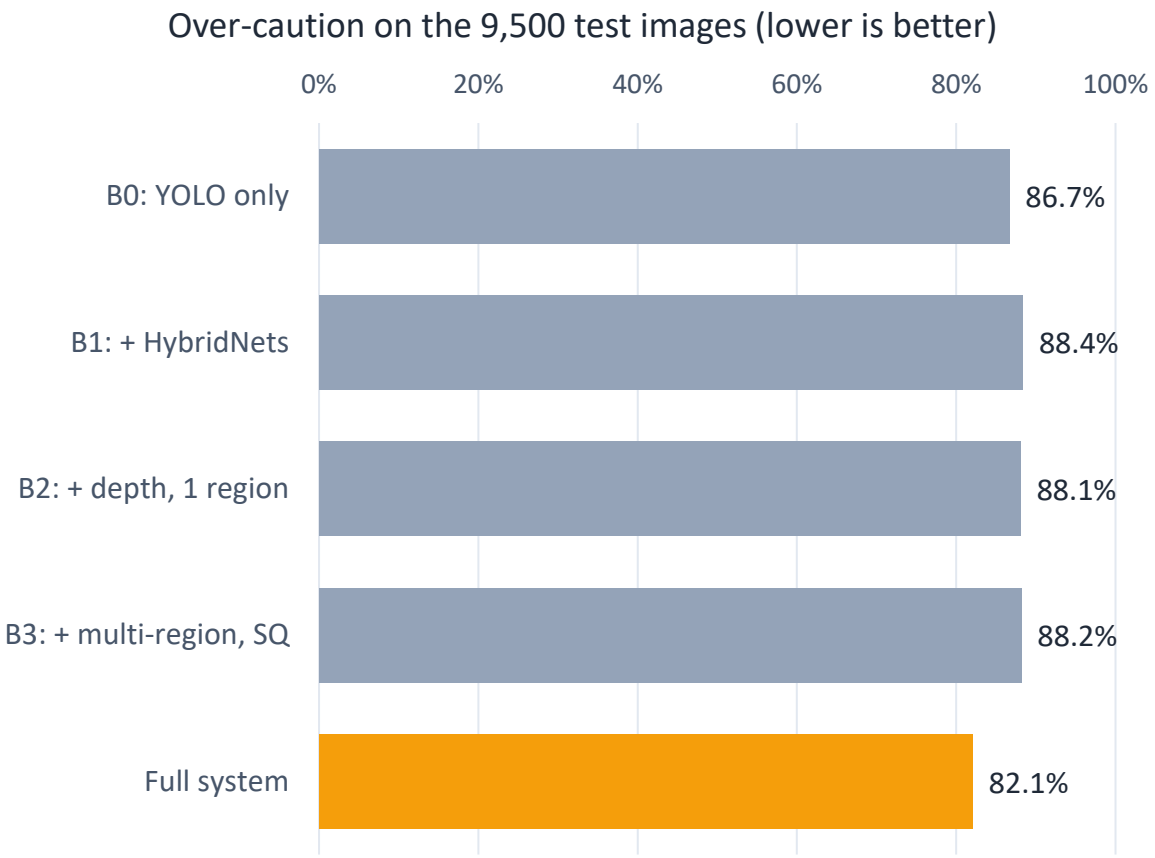
(d) failure
a distant parked car falls inside the corridor and is labeled PATH

(e) disagreement
the annotated ego lane spans the whole street

(f) truncation
an object cut by the image border is kept at CHECK

RESULTS

Each component, replayed on identical network outputs



Severity gates
STOP 3,893 → 1,166 and over-caution 5,157 → 4,799 images ($p < 0.001$); flips 12.2% → 8.3%

Full vs. YOLO-only B0
unsafe CLEAR 17 → 11 and over-caution 86.7% → 82.1%

Depth Anything V2
replacing it changes 3,631 advisories: per-image depth drives severity

Fallback
fires on 236 of 295 empty images; unsafe CLEAR 15 → 11 ($p = 0.12$, few events)

RESULTS

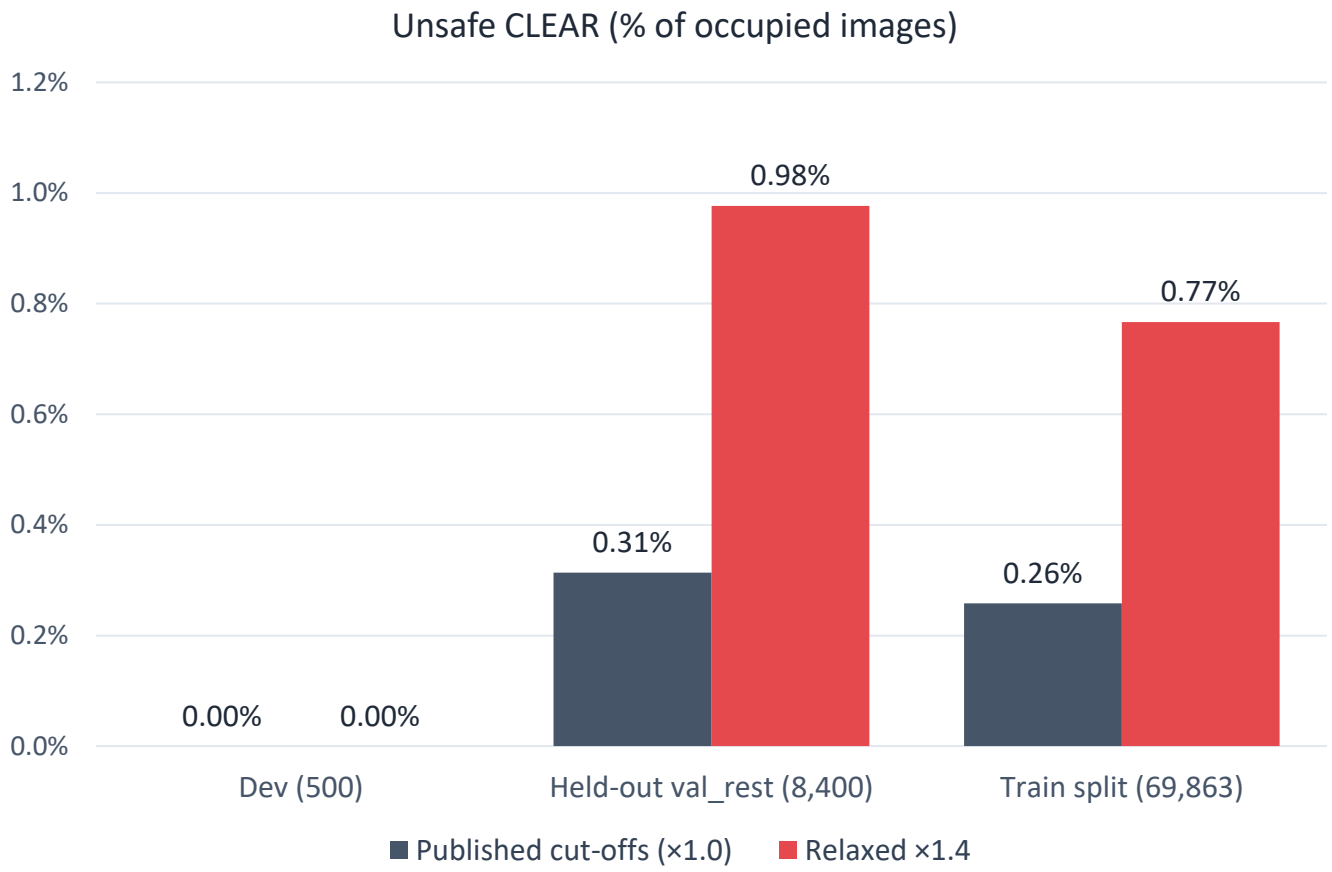
Removing one component at a time

Variant (test split)	Unsafe CLEAR	Over-caution %	BRAKE/STOP %	Advisories changed	McNemar p
Full system	**11**	**82.1**	**65.1**	–	–
without HybridNets (road prior)	13	80.3	65.6	364	0.62
without Depth Anything V2 (depth prior)	12	83.0	55.8	3,631	1.00
without multi-region (center only)	11	82.0	65.0	47	1.00
without scene quality	14	82.4	65.7	232	0.25
without severity gates	11	88.2	74.5	**4,024**	1.00
without depth-only fallback	15	82.1	65.1	236	0.12

Reading: the p-values test unsafe CLEAR, which has only about ten events, so none of these differences is significant on its own. The gates have the largest effect on the outputs (over-caution 5,157 vs 4,799 images, $p < 0.001$); depth drives the severity; HybridNets is under-used, since only 9.8% of corridor rows follow the detected lanes.

RESULTS

A margin chosen on the dev set does not transfer



0 → 0

unsafe CLEAR on the dev set at ×1.4, while over-caution falls 85.0% → 72.4%

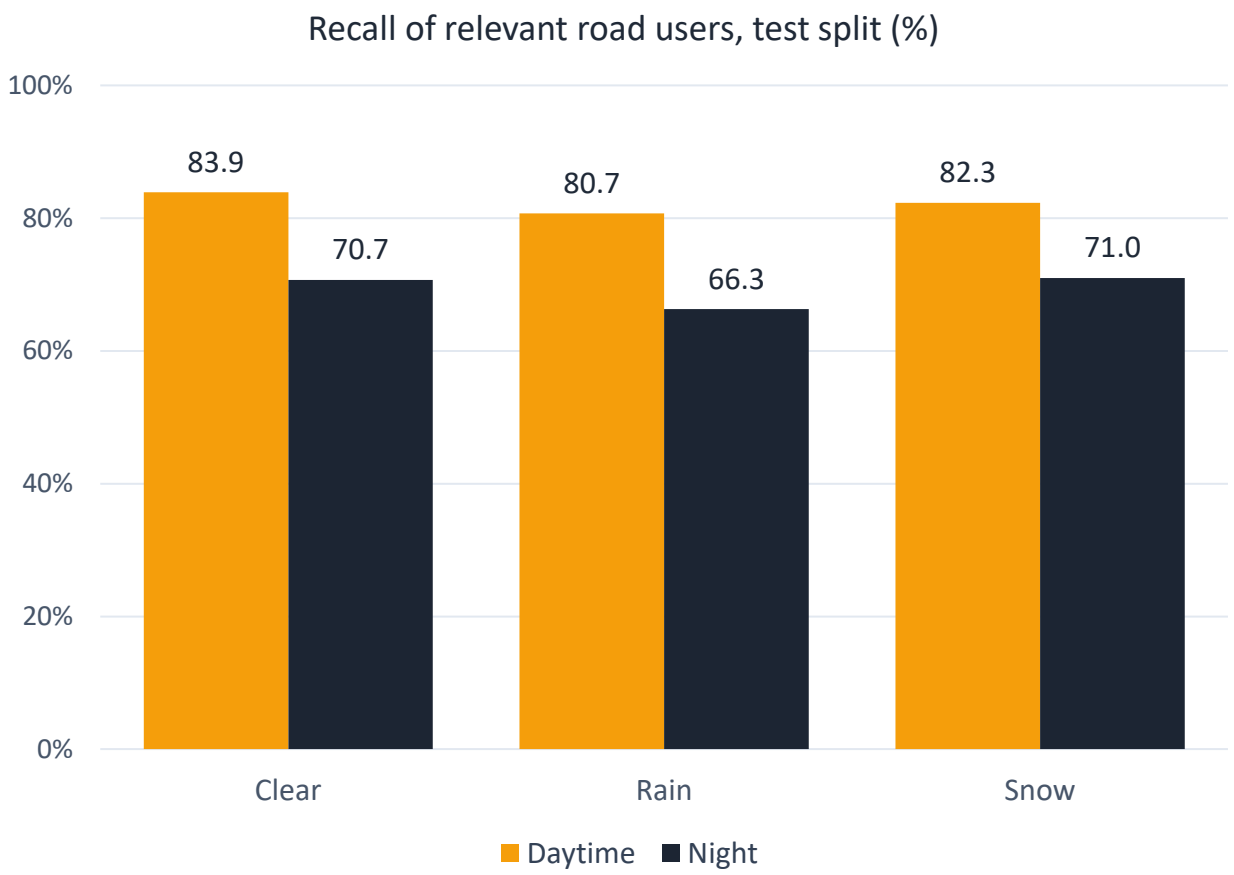
9 → 28

on 8,400 held-out images: unsafe CLEAR triples (0.31% → 0.98%)

The dev set has only 162 occupied images, too few to see a 0.3% event rate. The published cut-offs are kept.

RESULTS

Night, weather, perturbations and hardware



Night is the weak spot

Recall 70.4% vs 83.3% by day. Rain costs 3.2 points by day and 4.4 at night; snow almost none; fog: only 13 test images.

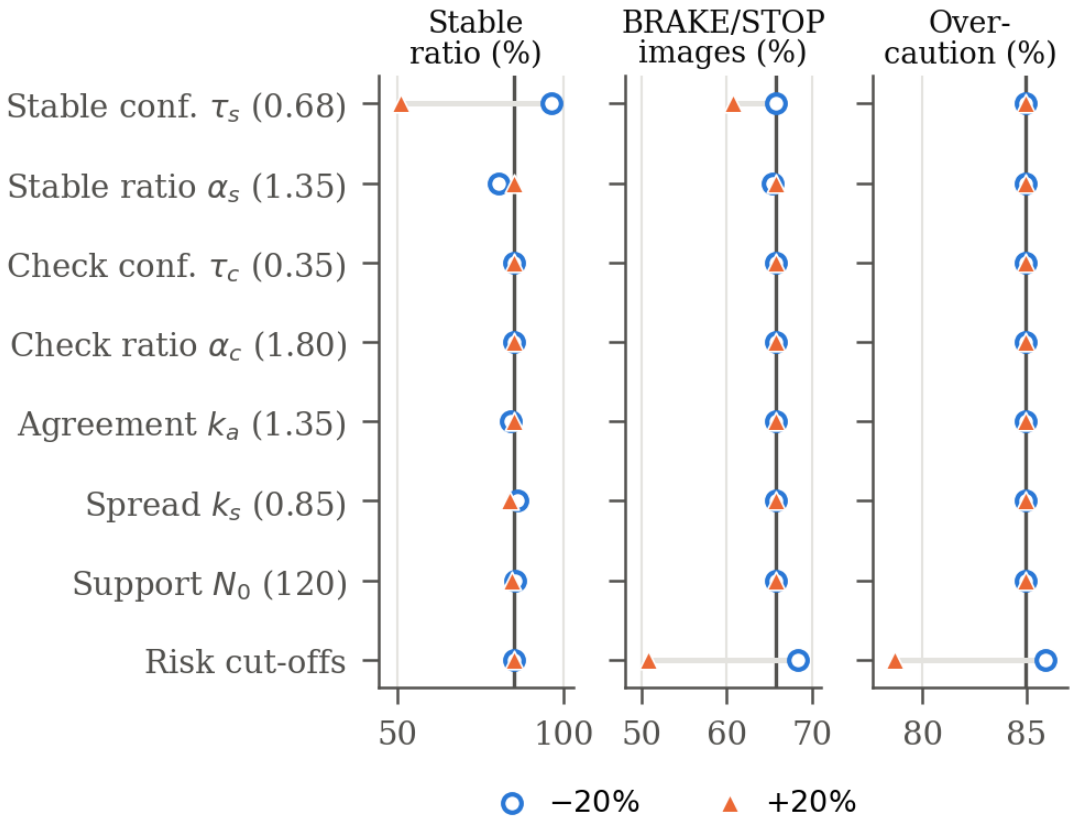
Mild perturbations

The advisory changes in 5.2% of image pairs (6.6% without gates).

Different GPU

RTX 3050 vs T4: 0 of 1,000 and 1 of 5,130 advisories differ.

Sensitivity of the fixed constants



≤ 34 points

change of the stable ratio when a reliability constant moves by ±20%, almost entirely through the STABLE threshold

≤ 5 points

change of the BRAKE/STOP rate; over-caution does not change

0 / 162

unsafe CLEAR on the occupied dev images, for every sensitivity setting

RESULTS

Supplementary check on 69,863 train images

Measure	Test, full	Test, no HybridNets	Train, full	Train, no HybridNets
Detection precision (%)	89.1	91.3	88.9	91.2
Recall, relevant road users (%)	78.9	76.0	79.4	76.6
PATH precision / recall (%)	9.2 / 53.8	9.9 / 54.4	9.1 / 53.3	9.8 / 54.0
Road IoU (HybridNets vs GT)	0.828	–	0.854	–
Unsafe CLEAR (% occupied)	0.34	0.40	**0.26**	0.40
Over-caution (% free lane)	82.1	80.3	80.8	79.1
BRAKE / STOP (% images)	65.1	65.6	64.9	65.4

Same picture on 7× more images. HybridNets was trained on these images, so its road output is slightly optimistic there (IoU 0.854 vs 0.828); the COCO-pretrained detector was not, and Depth Anything saw BDD100K images only without labels. Labels come from public copies that reproduce all 10,000 validation labels and masks exactly.

CLOSING

Limitations and next steps

Limitations

- Static images: no speed, time-to-collision or tracking
- Over-inclusive corridor: frequent over-caution
- Night recall and rare fog
- Lane relation approximated from drivable-area maps
- 2.3 s per image: an offline analysis tool

Next steps

Drivable-mask corridor: in a replay, over-caution falls 4,799 → 3,843 images with the same 11 unsafe CLEAR

Video: temporal consistency, speed and time-to-collision

Calibrated distance targets and human rating of the advisories

CLOSING

Every number can be regenerated

Released splits

Image-ID lists of every split with SHA-256 hashes (test prefix af51f530c5d9b3c5).

Byte-identical replay

Replaying the published constants on cached network outputs reproduces the original reports exactly.

Two GPUs agree

RTX 3050 and T4 with pinned library versions: 0 of 1,000 and 1 of 5,130 advisories differ.

Public code

Evaluation, tables and figures are generated by scripts; no number is typed by hand.

```
python scripts/download_weights.py
python run_paper.py input --output-dir outputs/paper
```

Takeaways

Conservative by design

11 unsafe CLEAR in 3,250 occupied test images

Reliability states are informative

non-STABLE estimates: 1.8× more false positives

The cost is measured and explained

over-caution comes from the corridor, with a tested remedy

Code, image-ID manifests and evaluation scripts

github.com/VuTien1311/roadsense-risk-assessment

Thank you · Questions?